

Class 8 – Arduino & Tinkercad

1. Start Smart – Observe

Have you ever seen lights blink in a pattern, like on a **Christmas tree** or during a festival?

How is this effect created?

- a. Someone turns the lights ON and OFF manually. ❌
- b. The lights are connected to a **microcontroller or circuit** that changes the patterns. ✅

Arduino

One popular tool used to create such patterns is **Arduino**.

- Arduino is a **small microcontroller** that can be programmed to control different electronic devices.
 - In this chapter, we learn how lights and other devices can be controlled automatically using Arduino.
 - We will learn how to:
 - Program Arduino.
 - Control electronic components.
 - Simulate projects using **Tinkercad**.
 - Create exciting and interactive circuits.
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2. Arduino

Arduino is an open-source electronics platform that is easy to use and is based on both **hardware and software**.

Arduino can:

- Read **inputs**, such as:
 - Light from a sensor.
 - A finger pressing a button.

- A message from a computer.
- These inputs are then changed into **outputs**, such as:
 - Activating a motor.
 - Turning ON a light.
 - Publishing information online.

Working of Arduino

Input → Process → Output

Examples:

- **Pressure → Input → Arduino → Output → Motor**
- **Smoke → Input → Arduino → Output → Smoke Alarm**
- **Temperature → Input → Arduino → Output → Thermostat**

Programming Arduino

- We control the Arduino board by giving it instructions using the **Arduino programming language**.
 - We write the code in the **Arduino Software (IDE)**.
 - The IDE helps us **create and upload programs** to the Arduino board.
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3. Arduino Hardware

Arduino boards are **small, programmable devices** used to create electronic projects.

Different Arduino boards provide different features for building:

- Sensors
- Robots
- Other interactive devices

Different Arduino Boards

Arduino UNO R3

- Arduino boards like **UNO**, **Leonardo** and **Micro** are the backbone of Arduino projects.
- They have been successful for many years.
- They continue to be a **reliable choice** for different projects.

Arduino Nano

- The Nano family includes compact and feature-packed boards.
- Examples:
 - Nano Every
 - Nano 33 BLE Sense
- They offer:
 - Bluetooth/Wi-Fi
 - Embedded sensors
 - Support for MicroPython
 - Support for Machine Learning

Arduino Mega 2560 Rev3

- The Mega family includes boards designed for projects requiring:
 - High computing power.
 - Numerous GPIO pins.
- They are suitable for complex projects such as:
 - Robotics
 - 3D printing
 - Home automation

Arduino MKR 1000 WiFi

- The MKR family has boards with built-in radio modules for communication such as:
 - Wi-Fi
 - Bluetooth
 - LoRa
 - They are powered by an **Arm Cortex-M0 processor**.
 - They also have a **crypto chip** for secure communication.
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4. Anatomy of an Arduino Board

Arduino has introduced many hardware designs over the years.

- Each design can have its own shape and form.
- However, some important components are present on **nearly every Arduino board**.

Important Components of Arduino UNO

Reset Button

- It restarts the Arduino.

USB Port

- It powers the Arduino.
- It allows us to **upload code**.

DC Port

- It powers the Arduino using an external source.
- It supports **7V to 12V**.

5V and GND Pins

- They provide **power and ground** for the circuit respectively.

Digital Pins (0–13)

- They are used to turn devices **ON or OFF**.
- Example: **LEDs or buzzers**.

ATmega Microcontroller

- It is the **brain of the Arduino**.
- It executes the code.
- It controls the digital and analog inputs/outputs.



Analog Pins (A0–A5)

- These pins read sensor values such as:
 - Light
 - Temperature
 - Sound
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5. Arduino Software (IDE)

IDE stands for **Integrated Development Environment**.

The **Arduino Software (IDE)** allows users to:

- Write code.
- Upload code to the Arduino board.
- Work without an internet connection.

Important Points

- It is especially useful for users with **limited or no internet access**.
- It is compatible with **any Arduino board**.
- It is a versatile tool for Arduino projects.

The Arduino IDE can be downloaded and installed from the **official Arduino website**.



Concept Check

Fill in the blank:

Digital pins on an Arduino Uno are labelled as:

0 to 13

They are used to **read or control simple ON/OFF signals**.

6. Tinkercad Circuit Design

Tinkercad is a **free, easy-to-use, cloud-based** tool by Autodesk.

It allows you to:

- Design circuits.
 - Simulate projects.
 - Include Arduino code.
 - Work without needing physical materials.
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7. Accessing Tinkercad

You can start working on projects and **save your progress online** by logging in to Tinkercad.

Steps:

1. Open a web browser.
2. Go to:

www.tinkercad.com

3. On the Tinkercad home page:
 - Click **Sign Up** if you are a new user.
 - Click **Log In** if you already have an account.
4. Enter your **email ID and password**.
5. The default screen will open.

Tinkercad has three main categories:

1. **3D Design**
 2. **Circuit**
 3. **Codeblocks**
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8. Tinkercad Circuit Window

To start designing a circuit:

1. On the default screen, select the **+ Create** drop-down button.
2. Click on the **Circuits** category.
3. The **Tinkercad Circuit Window** will open with different components.

Important Parts of the Circuit Window



Change Name

- Used to **rename your project**.



Code

- Gives access to the **code editor for Arduino programming**.



Start Simulation

- Simulates your code and circuit to **check their functionality**.



Send To

- Used to **share your project**.



Toolbar

Provides functions such as:

- Delete
- Rotate
- Undo
- Redo
- Wire colour
- Wire type

Search

- Used to **search for components**.

Components

- Area where you select electronic elements for circuit design.

Work Area

- Area where you **drag and drop components**.
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9. Lab Session – Creating Lighting Patterns with LEDs

Aim

Create a simple circuit with LEDs to produce a **sequential lighting pattern**.

In this pattern:

- LEDs turn **ON and OFF one by one**.
- They work at **set intervals**.
- This creates a **flowing visual effect**.

Starting the Project

1. Log in to **Tinkercad**.
 2. Go to the **Create** section.
 3. Select **Circuits**.
 4. Start a new project.
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10. Components Required

In the **Work Area**, drag the following components:

- 5 LEDs
 - 5 resistors (220 Ω)
 - Arduino UNO
 - 9V battery
 - Breadboard
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11. Connecting the LEDs

Step 1 – Select LED Colours

- Click on an LED.
- A **properties panel** will appear on the side.
- Select the desired colour, such as:
 - Red
 - Green
 - Orange
 - Blue
 - White

Step 2 – Insert LEDs and Resistors

- Insert the LEDs into the **breadboard**.
- Connect a **220 Ω resistor** from the **cathode of each LED** to the **GND rail** of the breadboard.

Step 3 – Connect GND

Connect the **GND pin of the Arduino** to the **GND rail** on the breadboard.

Step 4 – Connect 5V

Connect the **5V pin of the Arduino** to the **positive rail** on the breadboard.

Step 5 – Connect the LED Anodes

Connect the **anode** of each LED to these Arduino digital pins:

LED	Arduino Pin
 Red	Pin 3
 Green	Pin 4
 Orange	Pin 5
 Blue	Pin 6
 White	Pin 7

12. Programming the LEDs

Now open the **Code editor** in Tinkercad.

To Turn ON the LEDs in Sequence

1. Drag the **set pin to HIGH** block from the **Output** blocks under the **forever** block.
2. Click the number in the block.
3. Change the number to **3**.
4. Change the state to **HIGH**.

Why HIGH?

The red LED is connected to **pin 3**.

When the pin is set to **HIGH**, it supplies power to the LED.

 Therefore, the **LED turns ON**.

Add a Wait

- Add a **wait** block.
 - Set the time to **0.25 seconds**.
 - This pauses the execution of the program for a specified amount of time.
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Repeat for Other LEDs

Repeat the same steps for LEDs connected to:

- Pin 4
- Pin 5
- Pin 6
- Pin 7

For each LED:

Set pin to **HIGH** → Wait 0.25 seconds

13. Turning OFF the LEDs in Reverse Sequence

To turn OFF the LEDs, start with pin 7.

Sequence:

1. Set pin 7 → **LOW**
2. Wait **0.25 seconds**
3. Set pin 6 → **LOW**
4. Wait **0.25 seconds**
5. Set pin 5 → **LOW**
6. Wait **0.25 seconds**
7. Set pin 4 → **LOW**
8. Wait **0.25 seconds**
9. Set pin 3 → **LOW**
10. Wait **0.25 seconds**

What does **LOW** mean?

When the state is set to **LOW**, the pin stops supplying power to the LED.

→ Therefore, the **LED turns OFF**.

14. Test the Code

After completing the code:

1. Click **Start Simulation** in Tinkercad.
2. The LEDs will turn ON **one by one**.
3. There will be a **0.25-second interval** between each LED.
4. The sequence starts from **red and ends at white**.
5. Then the LEDs turn OFF in **reverse order**.
6. The OFF sequence starts from **white and ends at red**.
7. There is also a **0.25-second interval** between each action.

Complete Pattern

ON:

 Red → Green → Orange → Blue → White 

OFF:

 White → Blue → Orange → Green → Red 

15. Master Challenge

What happens when an Arduino pin is set to HIGH and LOW?

- **HIGH:** The pin supplies power to the LED, so the LED **turns ON**.
 - **LOW:** The pin stops supplying power to the LED, so the LED **turns OFF**.
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16. Breadboard

A **breadboard** is a tool used for building and testing electronic circuits **without soldering**.

Important Points

- It has **metal strips underneath**.
- These strips connect the holes in rows.

- **Power rails** are continuously connected for power distribution.
 - **Terminal strips** connect holes in each row.
 - Components placed in the same row can be **electrically connected**.
 - It makes it easy to **build and modify circuits without soldering**.
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17. Innovators – Rodney Brooks

Rodney Brooks is known for his pioneering work in **robotics**.

- He co-founded **iRobot**.
 - iRobot is the company behind the famous **Roomba vacuum robot**.
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Quick Revision

Arduino

- Open-source electronics platform.
- Based on **hardware and software**.
- Takes **inputs**, processes them and produces **outputs**.

Arduino UNO

- **Digital pins:** 0–13
- **Analog pins:** A0–A5
- **ATmega Microcontroller:** Brain of Arduino
- **USB Port:** Powers Arduino and uploads code
- **Reset Button:** Restarts Arduino
- **DC Port:** Supports 7V–12V
- **5V and GND:** Provide power and ground

Arduino IDE

- Used to **write and upload programs**.
- Can work **without an internet connection**.
- Compatible with **Arduino boards**.

Tinkercad

- Free, easy-to-use, cloud-based tool by **Autodesk**.
- Used to **design and simulate circuits**.
- Main categories:
 - 3D Design
 - Circuit
 - Codeblocks

LED Circuit

LEDs: 5

Resistors: $5 \times 220 \Omega$

Arduino: UNO

Battery: 9V

Board: Breadboard

LED Pins:

-  Red → **3**
-  Green → **4**
-  Orange → **5**
-  Blue → **6**
-  White → **7**

Remember

HIGH → LED ON

LOW → LED OFF

ON: Red → Green → Orange → Blue → White

OFF: White → Blue → Orange → Green → Red

Interval = 0.25 seconds